

# Bounded review: "Two Small-Order Classification Theorems for Signed Difference Sets" (Masselot, v1.0, 2026-08-12)

Reviewer: Fabian Arévalo (farev/Matematica), with substantial AI assistance (Claude). Every computational claim below comes from code committed under `masselot-review/` and can be rerun with the commands at the end. Review date: 2026-08-12. Scope, as requested: (i) theorem framing, (ii) completeness of the quotient reductions, (iii) earlier exact resolutions.

## Short version

Both theorems are correct. I re-derived every nonexistence claim with my own code, written from the definition alone, and checked all sixteen witnesses with a third validator that shares nothing with your two. Every count in Sections 5 to 7 of your note came out exactly the same. One extra result: the C18 quotient system of (36,29,4) is empty. That proves the C36 and C2 x C18 cases in one step, so your Corollary no longer needs the database entry it currently imports. Details below.

## What I checked, and how

My code (`qrlib.py`, `check_targets.py`, `verify_witnesses.py`) was written for this review, from the definition alone. The witness checker is my repository's `sdslib.check_sds`, written 2026-08-09 against Gordon's convention. None of it shares code with your two validators, with Gordon's `is_sds`, or with a SAT solver. All arithmetic is exact (Python integers; NumPy int64 only for batch filtering).

Before touching your claims I ran four known-answer controls. My chain code reproduces, set for set, the 40 labeled SDS(20,11,2,[20]) from my complete enumeration of 08-09. My two-layer code returns zero on SDS(36,29,20,[6,6]), matching my direct exhaust, and it keeps, rather than prunes, the branch of a known SDS(36,11,2,[6,6]) witness. The chain also returns zero on SDS(32,28,12,[32]), again matching my exhaust.

| your note says                                                                                                                                                                   | I get                                                                                                                                                                                                                                                           | match |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 16 valid witnesses, six of them the order-32 constructions                                                                                                                       | 16/16 pass <code>check_sds</code> ; the six order-32 file hashes match the table in your §4                                                                                                                                                                     | yes   |
| §5: 9,528 C8 vectors after sum and norm; 56 solutions; 5 affine orbits of sizes 8,16,8,16,8; 12 C16 survivors over orbit representatives; 248,832 final refinements; 0 solutions | same numbers; and with no orbit reduction at all: 144 C16 survivors over all 56 parents, 2,985,984 refinements, 0 solutions                                                                                                                                     | yes   |
| §6, C2 x C18: 144 candidates after canonical marginals and norm; 0 solutions                                                                                                     | 144 on the canonical pair; full enumeration over every marginal pair, no normalization: 0 solutions                                                                                                                                                             | yes   |
| §6, C3 x C12: 420 candidates; 0 solutions                                                                                                                                        | 420; full unnormalized enumeration: 0 solutions                                                                                                                                                                                                                 | yes   |
| §7: 106,353 C3^2 vectors after sum and norm; 9 projection solutions in one orbit; real side (7,2,2,2); one normalized instance; UNSAT by checked DRAT                            | 106,353; the 9 solutions are exactly the nine translates of (-3,2,...,2); the C2^2 system is exactly the four translates of (7,2,2,2); and a solver-free search over all 36 marginal pairs: 16,964,640 candidates, 0 pass the 35 correlation equations (43.7 s) | yes   |

| your note says                                                                                                     | I get                                                                                                                                                                             | match |
|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| census: the 68 Open entries of order $\leq 36$ at the frozen commit; 58 replications of farev/Matematica; 10 novel | my own 2026-08-09 fetch of the same commit has exactly those 68 Open cells; cell-by-cell diff: 58 agreements, 0 disagreements; your 10 novel cells are exactly the 10 I left open | yes   |

All of this runs in under a minute on a laptop. What I did not check: your CNF/DRAT chain (on purpose: for  $C_6 \times C_6$  my direct search replaces it rather than audits it), the contents of the Zenodo archive (I only checked the record and its hashes), and where your witnesses came from (which does not matter for whether they are valid).

## (i) Theorem framing

The framing is sound. Two things are better than what I usually see: the acceptance boundary in Section 2 (raw UNSAT is not evidence; witnesses must pass two validators with all normalizations forgotten), and the Corollary openly marking the one datum it imports instead of absorbing it silently.

Comments, none blocking:

1. Your definitions match Gordon's paper (his Lemma 1.1 is your equation (2); his  $|P| \geq |N|$  convention is your sign normalization) and the database checker. Your equation (1) is the same identity my census uses, so all three projects are provably about the same object.
2. Theorem 1 rests on two kinds of evidence: explicit witnesses on one side, an exhaustive refinement on the other. Sections 4 and 5 say this clearly, but the theorem will get quoted without them. One sentence in or next to the theorem statement would help.
3. The Corollary can become self-contained: see the observation below. Theorem 2 and the Corollary then merge into one clean statement: no abelian group of order 36 admits a signed (36,29,4) difference set.
4. Your Section 3 lemma has an ancestor worth citing: the sum and sum-of-squares parts of your equation (4) are Lemma 5.2 in Gordon's paper (intersection numbers; also standard in the design literature, e.g. Beth, Jungnickel and Lenz VI.5.4). Costs one line.
5. The evidence-type counts in Section 9 (16/13/4/9/14/12) match the machine-readable census exactly. I checked.
6. Two cosmetic nits. The file `sds_25_12_1_c5xc5.json` uses `group: "C_5 x C_5"` and `frozen_at_utc` where the other fifteen use `group_invariant_factors` (and one uses `frozen_date`); worth harmonizing before a journal submission. And your email called the source the "La Jolla covering repository", which is a different database; the note's name is the right one.
7. A remark for your discussion section, if you want it: at (32,20,4), existence splits exactly at cyclic versus noncyclic. In my census, (32,28,12) splits at containing  $Z_4 \times Z_4$  (it exists only in [4,8] and [2,4,4]). So no single subgroup rule explains order 32.

## (ii) Completeness of the quotient reductions

I tested this three ways: I checked your arguments as written, I re-proved the facts they depend on, and I reran every reduction with the symmetry arguments removed, so completeness no longer depends on them.

1. The projection lemma and its constants are right: kernel size  $m$  gives cells in  $[-m, m]$ , peak  $k + (m-1)\lambda$ , off-peak  $m\lambda$ . My systems use exactly these and get your counts, so we agree on the actual constraint systems, not just the prose.
2. Section 5 (C32). Your completeness argument needs three things. (a) The C8 enumeration is complete: confirmed, 9,528 and then 56 as complete counts from my own search. (b) The affine reduction is sound because the group actions lift: true. Translations lift along any surjection, and the unit maps  $(\mathbb{Z}/32)^x \rightarrow (\mathbb{Z}/16)^x \rightarrow (\mathbb{Z}/8)^x$  are onto, so multiplying by a unit of  $\mathbb{Z}8$  extends to an automorphism of C32. (c) Every survivor gets refined explicitly: confirmed, and also made irrelevant, because my rerun refines all 56 C8 solutions with no orbit reduction anywhere (144 C16 survivors, 2,985,984 refinements, zero solutions). Your ledger records a symmetry bug you caught and removed (quotienting a normalized slice by the full translation group). That is exactly the kind of bug reviewers fear in these ladders. The released pipeline does not have it, and the unreduced rerun confirms its output.
3. Section 6. The normalization step needs two facts the note uses without stating: the marginal systems have exactly the claimed solutions, and the two translation normalizations do not interfere. Both are true. Complete enumeration gives exactly  $\{(1,6,6)$  and its rotations $\}$  for C3,  $\{(4,9),(9,4)\}$  for C2, the four translates of  $(7,2,2,2)$  for  $C2^2$ , and the nine translates of  $(-3,2,\dots,2)$  for  $C3^2$ , with your 106,353 as the sum-and-norm count. Independence holds because the translation group splits by CRT into the two directions. My reruns then loop over every marginal pair instead of canonical ones, so the exclusions hold with no normalization argument at all. Your normalized counts (144 and 420) both reproduce.
4. Section 7 (C6 x C6). The reduction to one normalized instance is complete for the same two reasons, which I verified rather than assumed. Stronger: the SAT certificate no longer carries the result. A direct search over all 36 marginal pairs (16,964,640 candidates, exact integer filtering through the 35 correlation equations) finds nothing. Your DRAT proof and my search are fully independent, and they agree.
5. On trust, to be explicit: I did not audit your CNF encoder or your drat-trim run, and my own searches have their own trust surface (my code), controlled by the known-answer checks above. The two pipelines share no code, which is what makes the agreement meaningful.

Verdict on (ii): the reductions are complete as claimed, and the result survives with every symmetry argument stripped out.

### **(iii) Earlier exact resolutions**

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I know of none, and I am in a good position to say so: my repository censused the same 68 cells three days before you froze yours.

- Gordon's paper (read in full for this review): the sporadic table has no order-32 or order-36 rows, the only noncyclic SDS in it is  $(18,13,4)$  in  $\mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3$ , and his searches ran on cyclic groups. His database, at the commit you froze (still HEAD today), lists all seven  $(32,20,4)$  cells and the three noncyclic  $(36,29,4)$  cells as Open. The cyclic  $(36,29,4)$  "No" you import is his orbit exhaust.
- He, Chen and Ge (read in full): their families are  $(v, v-1, -1)$ ,  $(243,242,161)$ , a 3-group family  $(3^{2m+1}, 3^{2m+1}, 1)$ , and fourth-order cyclotomic parameters. Nothing at order 32 or 36 with your parameters.
- My repository (farev/Matematica, sessions of 2026-08-09): decided 58 of the 68 cells. The ten you call novel are exactly the ten I left open, and my README said so, with a cost estimate of 5.5

\*  $10^{11}$  nodes per group for (32,20,4). Your census agrees with all 58 of my decided cells, zero conflicts, now confirmed from my side as well.

- Web and literature searches (2024 to 2026, several phrasings, including the ternary-sequence and perfect-ternary-array literature, which lives at  $\lambda = 0$  and does not touch these parameters): nothing.

So, as far as two independent censuses and the field's two papers can tell, your ten entries have no earlier exact resolution. Your "novelty-supported, search is not proof" framing is the right level. One suggestion: Gordon maintains the database we both froze. An upstream report of your ten decisions (and, if you want, jointly with my audit of the database's stored witnesses, 147 of 280 of which fail the defining equation as exported) would put the novelty question in front of the one person most likely to know of unpublished prior work.

## An observation you may want for a revision

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The datum your Corollary imports can be proved inside your own framework in one line of computation: for (36,29,4), the C18 quotient system (kernel C2, cells in [-2,2], sum 13, norm 33, every nonzero shift 8) is empty. I verified this by complete enumeration (0.6 s; it also drops out of the C9 x C2 hierarchy: the C9 system has 9 solutions and none extends). An empty C18 system rules out every abelian group of order 36 with a C18 quotient, which means C36 and C2 x C18 at once. Combined with your Sections 6 and 7, the statement "no abelian group of order 36 admits a signed (36,29,4) difference set" then rests entirely on your own certified computations, and the database entry becomes a cross-check instead of a dependency. Take it with or without attribution; the code is in my repository under `conjectures/signed-difference-sets/masselot-review/`.

## Rerunning this review

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From `conjectures/signed-difference-sets/` at `farev/Matematica`:

```
python3 masselot-review/verify_witnesses.py    # 16/16 VALID, ~2 s
python3 masselot-review/validate_pipeline.py    # 4 controls, ~10 s
python3 masselot-review/check_targets.py        # all legs, ~55 s
```

Outputs land in `masselot-review/out/` (witness verdicts, control results, the per-leg report with every count above, and the run log). Pinned copies of your sixteen witness files, with hashes, are in `masselot-review/witnesses/` under your CC BY 4.0.